IBIToken

Deployment and on-chain operational evidence

Inteli | Module 6 | Group G01 | Official v2 delivery in preparation

Revision based on execution on 26 September 2026 (UTC). The internal technical revision is 4; the preceding official release remains v1. A new test deployment does not by itself constitute formal submission.

1. Purpose and evidence scope

This report connects economic rules, contract behavior and observed records. The delivered silent MP4 runs for approximately 2 minutes 40 seconds. Deployment and verification are public; the operational cycle was executed in Anvil. This distinction corrects the planned public purchases described in the supplied draft.

Environment	Proven by this evidence set
Sepolia / 11155111	Creation, deployment parameters, test-price confirmation and exact verification on Sourcify.
Anvil / 31337	Registration, fictional opening, purchases, restrictions, assessment, two claims and announced/cancelled recovery.
Off-chain portal	Documented functional boundary. The portal video cited in the draft was not supplied for this review; its timestamps are not treated as independently verified evidence.

Consolidated model

One issuance of 150 indivisible IBT; an initial reserve of 50 and a 20-token cap per registered person across linked wallets. Ordinary participant transfers are disabled. Eight semesters span four years from the recorded opening. Entitlement is proportional to units multiplied by holding time, against 15% of reported gross revenue. Under the accommodation model, one quota represents three nights; consumption is managed outside the blockchain.

Currency and identities are test data. tBRL is not real money. The demonstration price is 34,874.14 tBRL/IBT. An opaque ID neither proves identity nor guarantees anonymity: it depends on external registration and may be linked to other data.

2. Verified public deployment

The **01_publicar.js** script was used in Remix with human confirmation in MetaMask. Compilation: Solidity 0.8.34+commit.80d5c536, Osaka, optimizer 200, viaIR enabled and vendored OpenZeppelin 5.6.1.

`IBIToken: 0x021DD3c028FC0FFaA45F6D8cb0D0912fD1850520`

`ADMIN / treasury: 0x549852CA58C2e843e29428a81Ab97D2d613BB749`

`tBRL: 0xFEc48658BdaBAfbdB572204B632d4dfCCF1F89fE`

`Creation tx: 0x132a696441b2a2feae4e2ea4b233c90e2f4534afaff84c9d7064a8d84ac0ec4f`

Field	Verified result
Block / inclusion	11783256 / 2026-09-26 02:22:36 UTC
Status / gas used	1 (success) / 3,906,647
Effective gas price / fee	2.588339865 Gwei / 0.010111730168582655 SepoliaETH
Constructor arguments	admin = ADMIN; total emission = 150; validFrom = 0; validUntil = 0; stablecoin = tBRL.
Source verification	Sourcify: creationMatch = exact_match; runtimeMatch = exact_match.

Zero validity values indicate that opening had not been recorded in this public deployment. ADMIN and treasury initially share one address, while authority and inventory are separate concepts. No public USER1/USER2 account was used for purchases in this MP4.

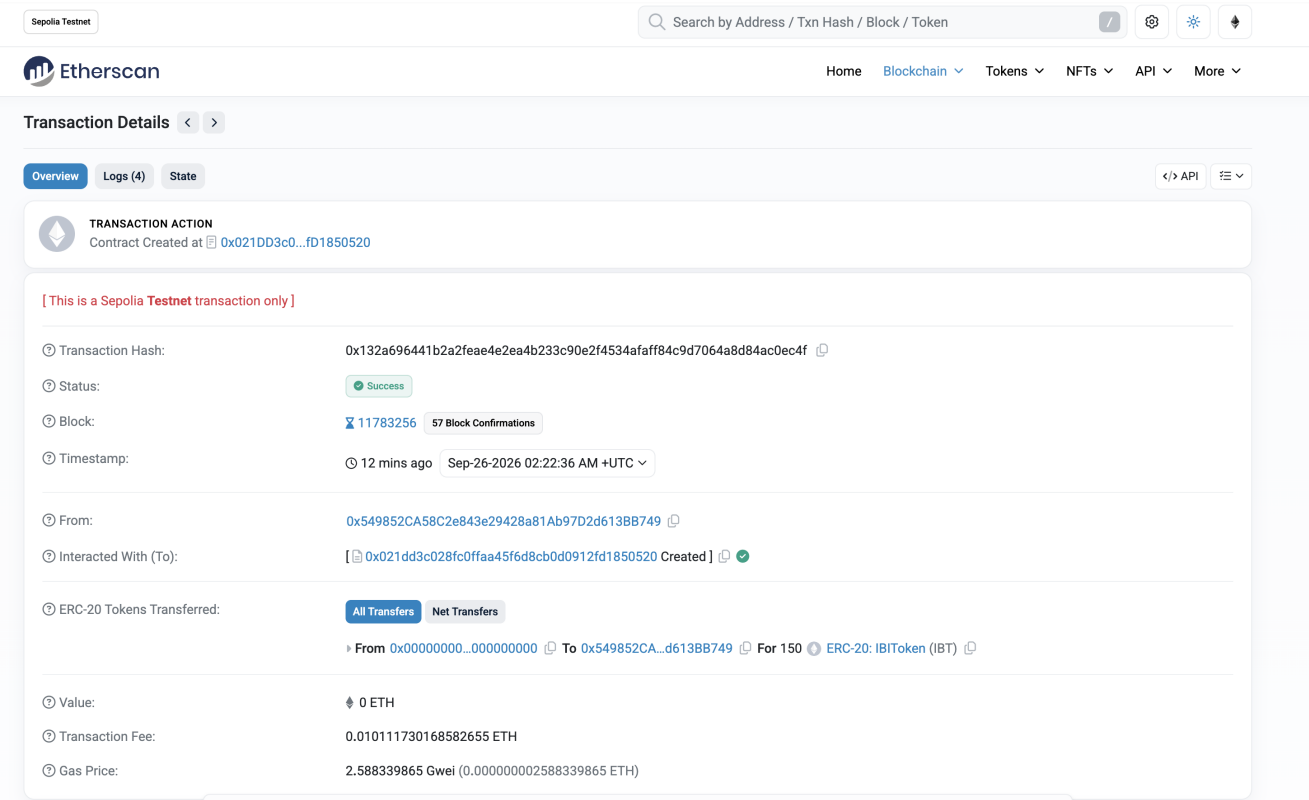


Figure 1. Public receipt: Success status, created contract and issuance of 150 IBT. Source: Etherscan, captured during this run.

Exact matching connects compilation to bytecode. It does not prove the memorandum is correct, that vulnerabilities are absent, or that external inputs are true. The fee above is gasUsed multiplied by effectiveGasPrice; it is not a guaranteed future cost.

3. Purchases and restrictions in Anvil

The academic console signs only in an isolated local instance, chain ID 31337. The accounts below are public test accounts, not three selected MetaMask accounts. Clicks produced real Anvil receipts; video cuts and frame holds support reading and dubbing.

```
ADMIN: 0xf39Fd6e51aad88F6F4ce6aB8827279cffFb92266
USER1: 0x70997970C51812dc3A010C7d01b50e0d17dc79C8
USER2: 0x3C44CdDdB6a900fa2b585dd299e03d12FA4293BC
```

Action	Observed
Register USER1 and USER2	WalletRegistered for distinct synthetic IDs. This does not prove real KYC.
Record opening	15 March 2027 00:00 UTC, fictional. First semester ends on 15 September 2027 00:00 UTC.
USER1: approve + buyPrimary	Buys 2 IBT at the start. Cost: 69,748.28 tBRL.
USER2: approve + buyPrimary	Buys 2 IBT at the exact midpoint. Same cost. Only Anvil time advances.
Unregistered purchase	WalletNotRegistered
Transfer USER1 to USER2	SecondaryTransfersDisabled
Premature reporting	PeriodNotClosed

After both purchases	IBT	tBRL
ADMIN / treasury	146	139,496.56
USER1	2	0
USER2	2	0

Operational interpretation

Each buyer received exactly the required test amount. Spending approval is a separate transaction. In buyPrimary, tBRL payment and IBT delivery happen together: a failure reverts both. The order reference prevents replaying the same purchase. This does not make the external accommodation ledger atomic or prove a PIX payment.

The three rejections above were checked through **eth_call**: they were not mined and have no transaction hash or paid gas. Disabling a secondary market blocks ordinary transfers while preserving controlled recovery. The person-level cap still relies on preventing one real person from receiving multiple different registry IDs.

4. Royalties and recovery: measured results

The test semester lasts **15,897,600 seconds**. USER1 accumulated 31,795,200 token-seconds; USER2 accumulated 15,897,600. Fictional revenue G = 150,000 tBRL; fund R = 22,500 tBRL.

R = floor(G × 1500 / 10000)
W = Σ(units × seconds held during the semester)
entitlement = floor(R × W / (150 × duration))

Holder	Due	Claimed	Final outstanding
USER1	300 tBRL	300 tBRL	0
USER2	150 tBRL	150 tBRL	0
Treasury	22,050 tBRL	0	22,050 tBRL

The deposit was 22,500 tBRL; 22,050 remained in the contract after the two claims. Treasury participates through inventory and reserve, receiving 98% in this example. Its currency balance after sales and funding is 116,996.56 tBRL. Claims did not consume participant IBT.

Rounding: the contract transfers the sum of individual entitlements (**totalDue**), which may be below R. A rounding difference is not deposited and remains with the funder. It was zero in this test. The memorandum should state the economic treatment of that difference.

Recovery	Result
requestRecovery	USER1 to a new empty wallet for the same person; request recorded.
reissue / eth_call	RecoveryNotReady; executeAfter = request + 604,800 seconds (7 days).
cancelRecovery	RecoveryCancelled. USER1 retains 2 IBT; no balance was transferred.

Execution after seven days was not performed. The delay provides a contestation window but does not automatically integrate support with the contract. The old holder or owner can cancel on-chain; the complete external workflow and independent review remain pending. The destination cannot be treasury, owner, pendingOwner or another registered person and must satisfy the remaining code restrictions.

5. Architecture, operation and critical analysis

Account, custody and accommodation

The portal account brings together application, approval, wallet binding, orders and quotas. Names, contacts and documents remain in external systems; the contract receives opaque IDs. Approving an application grants neither tokens nor stays. A purchase recognized by the indexer updates rights in the external ledger. Using a quota does not burn IBT or, by itself, reduce royalties. Cancellations remain manual and provisional.

Three custody modes are envisaged: IBITI custody, assisted self-custody and direct self-custody. The automated institutional demonstration path works only through the Anvil adapter. It does not represent production key custody, real payment or a contracted service. Portal separation between supervision and finance is application authorization; it is not a contract-enforced multisignature requirement.

Finality and persistence

A receipt with status 1 proves successful inclusion, not finality by itself. Reorganizations may occur before finalization. The indexer design uses finalized state before making quotas available, introducing a delay after purchase. The transaction is not atomic with the off-chain database; idempotency, failure handling and reconciliation remain necessary.

Verifiability and concentrated authority

Events record opening, registration, purchase, reporting, claims and recovery. Receipts and bytecode support independent verification when the correct network is specified. Local operations do not appear on Etherscan. The owner remains a single authority: it reports revenue, registers wallets and opening, and can pause operations including claims. Two-step ownership transfer does not automatically move treasury or remove compromised-key risk.

Cost, limits and economic interpretation

The public deployment cost is measured in section 2; purchase costs in this collection are local and must not be presented as fees paid on Sepolia. Reporting iterates through tracked participants and cost depends on state. This report claims no public benchmark with 150 holders. Fifteen percent of revenue is the allocation base, not a guaranteed 15% return on the purchase price.

The contract does not verify actual revenue, physical opening or document authenticity. A hash can establish integrity against a preserved file; it does not validate its content. Civil identity data should not be placed on a permanent public network.

6. Evolution, evidence and remaining work

Change from v1	Rationale and evidence
Manual delivery → paid purchase	Payment and delivery together; local receipts and PrimaryPayment/PrimaryPurchase events.
Wallet cap → person cap	Closes concentration across linked wallets; verified code. Two wallets belonging to the same person were not exercised in this video.
Balance snapshot → holding time	Avoids full-semester entitlement for late entry; measured 300/150 amounts.
Fixed dates → recorded opening	Calendar tied to operation; premature reporting rejected locally.
Secondary transfers → disabled	Consolidated project decision; local rejection even between registered holders.
Immediate recovery → 7 days	Traceable request and cancellation; early execution rejected.

Remaining work: shared pillar governance, 2-of-3 treasury, selective pausing and integrated support/contestation; production KYC and custody; external substantiation of revenue and opening. Completed recovery after seven days, ownership transfer and multiple wallets for one person were not filmed in this MP4. Public purchases and a complete public cycle remain unevidenced in this collection.

Verifiable index

The official artifact contains only this PDF. The MP4, narration and JSON records are working material under **output/video/ibitoken-v2/** in the repository. The table below identifies the main records.

```
buyPrimary USER1 | Anvil | block 11 | gas 314737
0xb1ff3a0fa3e04bfa28a736439e7977c90c2b5c4e2d7cd5a69a3c4027ce24532a

buyPrimary USER2 | Anvil | block 14 | gas 320927
0xdfb961ffb0f6ea9f0cb012498a06cf6499097b655ba09c9b6312dc410c4ded7b

reportRevenue | Anvil | block 17 | gas 344717
0x5c55d3c71d200bf87dafeed1914534ac368620e5c203a525b18ee4de96b90268

claimRoyalty USER1 | Anvil | block 18 | gas 92949
0xbce186b1b9e77f4c287aef67a444bb93f873406ad7e14be23872debb5265e46a

claimRoyalty USER2 | Anvil | block 19 | gas 92949
0x5ba6d23669af16fd55069f2439bcf2a60753242fb9bc431f9771d595bb6cb343

cancelRecovery | Anvil | block 21 | gas 41586
0x4839d0894b51b8d2ab0b6261b9289400e06d02e3cef0d94d44991f37e87b5b18
```

Etherscan: public deployment receipt | Sourcify: exact_match and source

Conclusion: the evidence supports the revised implementation and its local cycle without confusing an academic simulation with complete public operation. This revision corrects official version labeling, replaces empty fields with real records and declares what remains undemonstrated.